We would like to sincerely thank the four anonymous Reviewers for providing feedback and constructive comments that helped us clarify and improve the quality of our paper. Below, we would like to highlight the most significant changes we have made in response to the reviewer comments:

- We have conducted 12 more sets of experiments (depths = {50x, 100x, 150x, 200x} and 3 runs for each; 4*3 = 12) for random sampling on *B. divergens MO1* as a part of the preliminary analysis.
- We have split Figure 1 into two distinct Figures to reflect the results based on fixed and variable coverage depths separately.
- Conducted 4 additional sets of experiments to trace the runtime and memory usage of AWin*K* and added a separate section (Section 4.5) in the manuscript.
- Updated the description of Steps C and D of AWin*K*'s algorithm.
- Added another supplemental table based on the analysis of unassembled contigs for assemblies at higher coverage depths of *C. elegans*.
- Added NG50 and NGA50 scores as supplemental.
- Added supplemental tables with numerical assembly statistics.
- Added citations to the first and second paragraphs.
- Updated the source code using main functions where applicable.

Reviewer 9ERo:

Major

1. The main motivation for this work is that existing assemblers (e.g., hifiasm and HiCANU) cannot run on ultra-deep sequencing data (e.g., they run for days and crash due to lack of memory). However, the authors did not provide the runtime and memory consumption of their method.

   Answer: We added a section on AWin*K*'s runtime and memory consumption in the revised version.

2. What parameters do the authors recommend to use?

   Answer: It is not clear what parameters the reviewer is referring to. If the reviewer is asking about Hifiasm or HiCANU, we've used the default parameters. AWinK has four parameters: (i) length of uni*k*mers, *k* (in Step A), (ii) *k*-mers in the interval [μ−tσ,μ+ tσ] of the *k*-mer distribution considered uni*k*mers, *t* (in Step A), (iii) scaling factor, *s* (in Step C), (iv) target coverage, D_t (in Step D). We have used *k* = 21 and *t* = 3 as recommended by [doi:10.1101/gr.279308.124]. Regarding the scaling factor *s*, we have used *s*=10,000 for *C. elegans* and *s*=1,000 for *B. divergens MO1.* For the target coverage D_t, we have tested D_t between 20x and 200x. Based on our experiments, we recommend using values between 30x and 100x for D_t.

3. I would also like to see what proportion of the minimum tiling path is actually chosen by AWinK (of course, this is hard on real data, but should be possible on synthetic one).

Answer: Given that the depth of coverage is usually much higher than 1-2X, it is very likely that the minimum tiling path is not unique, i.e. it is likely that there are several tiling paths that use the same number of reads. Because of this, determining the number of reads chosen by AWinK that are on a specific MTP would not be very informative.

Minor

1. I find figures like Figure 3 hard to read, e.g., comparing the oracle 10x to the reference. Having the raw number in supplementary materials might help.

   Answer: We added Supplemental Tables for the numerical data used in all the figures

2. Figure 8 (right): It would be nice to have more data between 50x and 100x, at least until the number of contigs goes up. As it is now, it is hard to see where the minimum number of contigs is attained.

   Answer: We added Supplemental Table 2 for the other depths

3. I would like to see more explanations for what happens, e.g. why does AWinK produces more contigs than the other methods for the same coverage in Figure 10 (right).

   Answer: We noticed that too. We determined that the vast majority of those extra contigs are unassembled reads as explained in lines 690-695.

4. I find figures like Figure 11 hard to read as well; the font is too small.

   Answer: Figure 11 was generated using the BioNano pipeline as a PNG, so it would be hard to edit.

5. 535: "First, AWinK creates a new bin and it assigns reads with $c \geq 6$ to it until the target coverage ratio D/s is reached." It seems to me that the order here matters, but it is not explicit. Explicitly state you take the reads with an increasing unikmer count rather than state "next unikmer count" at the end of the paragraph.

   Answer: We are not sure we understand the question completely, but we revised the description of that step since it was not clear to the other reviewers as well.

6. 536: Use c > 5 to stay consistent with line 532?

   Answer: Done.

7. Figure 6: The spike is hard to see. Maybe providing a zoomed version at the top right of the figure would help.

   Answer: We have put a zoomed in version in the revised version.

8. Table 2: AWinK manages to reach an identity of 100.025 %. This is likely a typo or a rounding issue.

   Answer: It was a typo and we corrected it in the revised version.

9. Nitpicking implementation: When writing a Python script, I'd recommend writing a main function, which would be called after the test if **name** == "**main**". Not writing such a function (i.e., simply writing code after the test) is a potential footgun, as it implicitly creates global variables. See below for a naive version of the problem.

```
def f():
    print(a)  # expecting "NameError: name 'a' is not defined"


if __name__ == "__main__":
    a = 1  # actually creates a global variable, silencing the
error above
    f()
```

Answer: We have updated the source code accordingly.

Reviewer UrEJ:

Major

- Dataset choice is very limited

The paper evaluates only two datasets, both from relatively small genomes. This narrow scope weakens the conclusions drawn from the experiments (some of these concerns are expanded below). It would be valuable to test the method on more challenging, repeat-rich genomes and regions (e.g., a human genome, particularly its highly repetitive centromeric regions). The T2T CHM13 reference genome and its HiFi reads are available and would be a good starting point.

> **Answer**: Due to the cost of generating PacBio data, ultra-deep sequencing data (>1000x) for large genomes (like human or mouse) is not realistic. We expect that ultra-deep would be only available for small genomes (i.e., less than 100Mb).

- Results are not very promising

Although the paper shows that AWinK improves genome fraction and sequence identity relative to other selection strategies, the results are problematic and not very promising for four main reasons:

1. These observations are from a single dataset only. A diverse range of datasets is necessary to substantiate claims about the method's strengths (as also discussed above).

   Answer: We agree that it will make the study more convincing if we had more data sets, but that there are not many ultra-deep HiFi datasets available. It is also the reason we are the first to report on the weakness of these popular assemblers with high coverage HiFi data.

2. Assemblies generated from AWinK-selected reads are highly fragmented (Figure 10). The high sequence identity might simply reflect the large number of short contigs. This requires justification. Using NA50 or NGA50 is probably better to justify such correlations.

   Answer: We looked into the highly fragmented assemblies from AWinK-selected reads. Our new Supplemental Table 4 shows the length distribution of these short contigs (length < 50kb and length < 100kb). We observed that the majority of these short contigs are unassembled HiFi reads. To avoid over-inflating the sequence identity metric, we recalculated it considering only the contigs with length >= 100 kb from each assembly.

3. Random sampling performs almost as well as AWinK, and at 100x coverage even slightly better, in terms of genome fraction. If simple random downsampling generates comparable results, it is unclear why one should generate ultra-deep datasets in the first place. The paper needs further discussion to justify the value of ultra-deep sequencing.

   Answer: When users choose to sequence a genome with PacBio technology, they usually purchase one or more SMRT flow cells. Each flow cell generates a fixed amount of data/reads. As a consequence, it is common to get a lot of PacBio reads (i.e., very high coverage) when the genome is small. One could use multiplexing to use a portion of a flow cell, but that comes with other technical issues.

4. Table 2 shows a sequence identity greater than 100% for AWinK at 100x coverage. Why is this the case?

   Answer: It was a typo and we correct it if allowed to revise.

- Lack of results to support observations

The paper makes several additional critical observations without adequate supporting evidence, mainly in three places:

1. Figure 1 gives multiple messages. If the main point is that higher coverage does not yield better results, the figure should be designed accordingly. Currently, coverage is fixed for random sampling but varied for other criteria, which appears arbitrary.

   Answer: Good point. We have now split Figure 1 into two separate figures accordingly. We conducted an additional 12 sets of experiments for random sampling at four different coverages, {50x, 100x, 150x, 200x}, with three independent runs for each.

2. Section 3.2 claims that certain selection strategies produce incorrect assemblies (lines 354–357, left column) based on contig length. Contig length alone is not a reliable indicator of assembly correctness, and, in fact, contigs 2–6 appear to match their reference lengths.

   Answer: We agree that obtaining the correct contig size is necessary for a good assembly, but not sufficient. Therefore, for alignment based quality assessment, we also provided dotplots in Figure 4.

3. Several important claims are made in the first two paragraphs of the Introduction. Please cite appropriate sources to support these statements instead of assuming they are common knowledge.

   Answer: We added the following references in the first paragraph

   - doi: 10.1101/gr.279308.12
   - doi: 10.48130/gcomm-0025-0015

   where the authors discussed various challenges in *de novo* genome assembly

In the second paragraph we added references to

- The HiCanu paper (doi:10.1101/gr.263566.120) shows that with ~30× HiFi data, assemblies of diploid human genomes achieved “*superior accuracy and allele recovery.*”
- Yu *et al.* evaluated multiple HiFi assemblers at 10×, 20×, 30×, 50×; contiguity and completeness improved steeply up to ~20× and then changed little from 20×→50×, indicating 30×–50× is an effective range for strong assemblies on tested genomes. (doi: 10.1101/gr.278232.123)

- Missing experiments

For real datasets, comparing the generated assemblies to an existing reference genome is necessary to see the metrics such as genome fraction, identity, NGA50 and NA50.

Answer: For real HiFi datasets, often there is no reference genome available, like the case of the *Babesia MO1 divergens* in our paper. On synthetic HiFi reads, we added Supplemental Figure 1 and Supplemental Table 3 that show the NGA50 and NA50.

I suggest adding results on how the selection strategy impacts the overall runtime and peak memory usage on the assembler. Simpler genome graphs may lead to better speed and memory usage.

Answer: We added a section on AWinK’s runtime and memory consumption in the revised version.

Minor

– The authors do not justify the use of minimap2 to measure sequence identity. QUAST or dnadiff are more commonly used for this purpose.

Answer: We believe that this choice is justified because Minimap2 is the default aligner used in QUAST

– In Section 3.4 (“D. Select Reads”), the algorithm is not described clearly. After starting with the high-frequency bin, how does one iterate over the remaining bins? The procedure should be stated explicitly rather than left to inference.

Answer: We have revised the description of step D for better clarity

Reviewer xzND:

1. However, as to results presented in Figure 10 as well as Table 2 - it seems like AWinK also shows quite some variance and simpler strategies yield better results on low-coverage datasets. As per Table 2 - AWinK also shows noticeably lower sequence identity ~1.6% and just barely higher genome coverage ~0.08% on 50x. Based on Figure 9 is looks like the optimal coverage is somewhere between 60x and 100x, but the Table 2 doesn't show exact values. The authors also do not present any information about misassemblies, which for me is a very important property.

Answer: We agree that we could have looked into misassemblies. However, we show alignment results for oracle approach via dotplots in Figure 4.

2. I am also confused by sequence identity exceeding 100% for 100x.

Answer: We corrected the typo.

3. Thus, it remains completely unclear why one should prefer AWinK over other strategies. As the authors do not provide any data about running time or RAM consumption, it's not clear how useful this tool in practice.

Answer: We added a section on AWinK's runtime and memory consumption in the revised version.

4. I would also like to see the tool evaluated on larger and more complicated genomes, e.g. human or mouse.

Answer: Due to the cost of generating PacBio data, ultra-deep sequencing data (>1000x) for large genomes (like human or mouse) is not realistic. We expect that ultra-deep would be only available for small genomes (i.e., less than 100Mb).

5. On a minor side, the authors use term "barcode", which in my opinion has a fixed meaning in genomics and transcriptomics and does not match the term used in the paper.

Answer: We agree that in genomics the term "barcode" is used to denote the process of tagging DNA. We will consider the term "signature" instead of "barcode" in the expanded (journal) version of the manuscript.

Reviewer 7dKB:

1. However, I'm not fully convinced about the accuracy of the proposed approach, and I do not believe that Figures 8, 9, and 10 effectively illustrate the method's potential. For example, in Figure 8, although the contig lengths at 100× coverage are close to those of the reference, the number of contigs is substantially higher. This makes it difficult to determine the optimal coverage for this case. It would be more informative if the authors provided a combined metric or summary that better captures the trade-offs across multiple assembly quality dimensions.

   Answer: New Supplemental Figure 1 and Supplemental Table 3 now show the NG50 and NGA50. Like genome fraction and sequence identity, we see a similar trend for NG50 and NGA50 scores as well where the assembly quality peaks between 50x and 100x coverage depths.

In addition, Figure 10 is difficult to interpret, as the contig lengths are very similar across most approaches. Clarifying the visualization would significantly improve readability and help the reader understand which setting yields optimal results.

I also have the following additional concerns:

- The statement, "It is common knowledge that a sequencing depth of 30–50× is ideal to obtain a complete and highly contiguous assembly from PacBio HiFi reads," is presented without citation. A reference should be provided to support this claim.

  Answer: The HiCanu paper (doi: 10.1101/gr.263566.120) shows that with ~30× HiFi data, assemblies of diploid human genomes achieved "*superior accuracy and allele recovery.*"

  Yu et al. evaluated multiple HiFi assemblers at 10×, 20×, 30×, 50×; contiguity and completeness improved steeply up to ~20× and then changed little from 20×→50×, indicating 30×–50× is an effective range for strong assemblies on tested genomes. (doi: 10.1101/gr.278232.123)

  We added references to both papers in the revised version of the manuscript.

- Verkko is widely considered one of the most reliable assemblers for long-read data. I was surprised that the authors did not evaluate it when analyzing the initial 3000× dataset. Moreover, Verkko is not cited anywhere in the manuscript, which seems like a notable omission given its relevance.

  Answer: We agree that Verkko is a good assembler when one has both HiFi and Oxford nanopore reads. But when given only HiFi reads in input, it underperforms [see, e.g., doi:10.1101/gr.278232.123]. For this reason, we did not consider Verkko in our study.

- The parameters used with hifiasm are not provided and should be included for reproducibility.

  Answer: We have used the default parameters and will mention that in the manuscript if allowed to revise.

- In the preliminary analysis, the authors suggest that metrics such as the lengths of the three largest contigs and the total number of contigs are sufficient to support the claim that "more HiFi data does not lead to better assemblies." While these metrics offer a basic indication of assembly quality, they are not comprehensive. More robust and widely accepted evaluation methods—such as reference-based metrics like NG50, BUSCO, and others—should be employed before making such a broad claim. This concern also applies to the final analysis, where similar conclusions are drawn without sufficient support from established quality assessment tools.

  Answer: Our real-data analysis is based on *B. divergens MO1* for which a high-quality T2T reference genome is not yet available. Therefore, it is not possible to provide alignment based quality metrics with respect to the reference genome, because it does not exist. However, we have added NG50 and NGA50 scores for our experiments on synthetic datasets generated from the reference genome of *C. elegans* in Supplemental Figure 1 and Supplemental Table 3.